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The Assisted Climb: Keeping Learners in the Pilot Seat in the Age of Generative AI

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The Assisted Climb: Keeping Learners in the Pilot Seat in the Age of Generative AI

Abstract

Agentic AI represents a fundamental shift in educational technology, moving beyond AI as a reactive tool towards AI as an active participant in learning. As systems increasingly exhibit interactivity, bounded autonomy, and adaptability, they redistribute epistemic labour through problem framing, knowledge generation, and evaluation, requiring pedagogical models that preserve learner agency. This paper introduces the Assisted Climb, a conceptual framework for human–AI co-agency that reconceptualises Nottingham's Learning Pit for learning environments characterised by artificial agency. Rather than viewing AI as a source of answers, the framework positions learning as a co-regulated process in which humans and AI contribute differently to knowledge construction while learners retain epistemic responsibility.

The model comprises three regulatory functions: Map, Torch, and Compass, that scaffold conceptual and epistemic conflict, enabling learners to critically engage with AI-generated contributions without relinquishing ownership of inquiry. A mixed-methods pilot study involving 41 Year 9 students explored the framework through surveys, ethnographic observations, semi-structured interviews, and student workbooks. Findings demonstrate how the Assisted Climb supports productive human–AI co-agency by helping learners negotiate epistemic labour, regulate AI participation, and sustain conceptual struggle. The paper contributes a pedagogical framework for learning with agentic AI while preserving human epistemic agency.

Key Words: Agentic AI, Human-AI co-agency, Epistemic agency, Epistemic labour

Practitioner Notes

What is already known about this topic

- Agentic AI is reshaping learning by sharing cognitive work with learners.
- Existing pedagogies were not designed for AI as a learning partner.
- New approaches are needed to preserve learner agency.

What this paper adds

- Introduces the Assisted Climb framework for human–AI co-agency.
- Demonstrates improved critical and reflective AI use among learners.
- Proposes Generative AI meta-literacy to regulate human–AI learning.

Implications for practice and/or policy

- Teach learners to question and verify AI outputs.
- Embed human–AI co-agency within AI literacy curricula.
- Design learning that keeps learners in the "pilot seat”.

AI usage declaration

The authors confirm ChatGPT was used to assist in the creation of Figure 2, 3 and 4 based on the original ideas and prompts written by the authors. All three outputs were analysed and redrafted until the authors agreed they were an accurate representation of their own original ideas.

Introduction

For many educators, *The Learning Pit* (Nottingham, 2020), shown in Figure 1, has become a widely adopted framework for helping learners understand the emotional and cognitive processes involved in learning. By legitimising confusion, uncertainty, and productive struggle, it frames learning as movement from initial confidence, through conceptual conflict, towards deeper understanding (Nottingham, 2024). This conception aligns with Piaget's (1952) account of cognitive conflict as the catalyst for conceptual change, Vygotsky's (1978) emphasis on socially mediated learning within the Zone of Proximal Development, and broader theories that position disruption and reflection as central to knowledge construction (Kitchener, 1983).

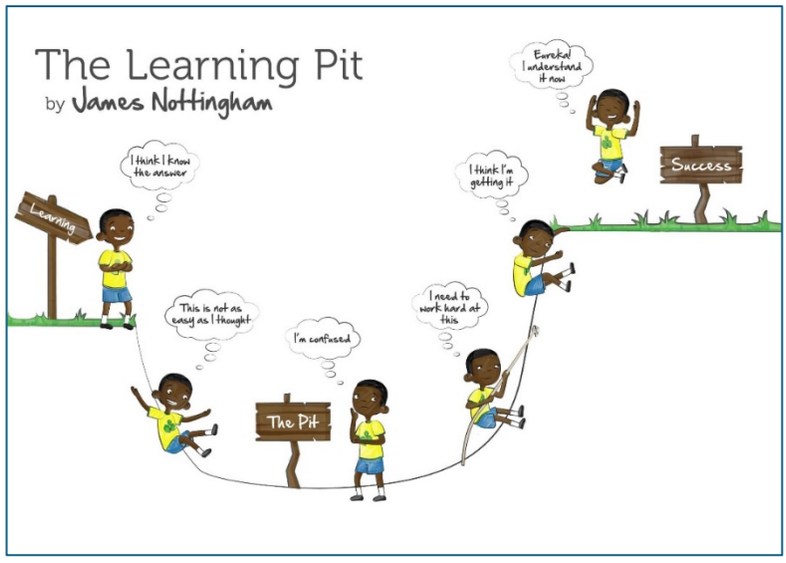


Figure 1. The Learning Pit. (Nottingham, 2020)

The emergence of agentic AI fundamentally alters the conditions under which this model operates (Adel, 2024). Unlike earlier educational technologies that primarily responded to user inputs, increasingly agentic AI systems exhibit bounded autonomy, adaptability, and interactive decision-making, enabling them to contribute to problem framing, knowledge generation, and evaluation alongside learners (Yadav & Shrawankar, 2025). It could therefore be argued AI is no longer simply a tool for accessing information, but an active participant in learning, redistributing epistemic labour across human and artificial agents (Yusuf et al., 2024). This shift challenges traditional assumptions about who performs the intellectual work of learning and how understanding is constructed (Jovanović & Campbell, 2022).

Within this context, as shown in Figure 2, the Learning Pit becomes increasingly complex because while AI is often presented as a scaffold that reduces barriers to learning (Zhai, 2022), its agentic capabilities can instead redistribute the cognitive and metacognitive work traditionally undertaken by learners. Rather than acting solely as a bridge across difficulty (Nottingham & Nottingham, 2017), AI may function more like a travelator, advancing learners through tasks while reducing opportunities for the conceptual struggle associated with Piagetian equilibration (Fan et al., 2025).

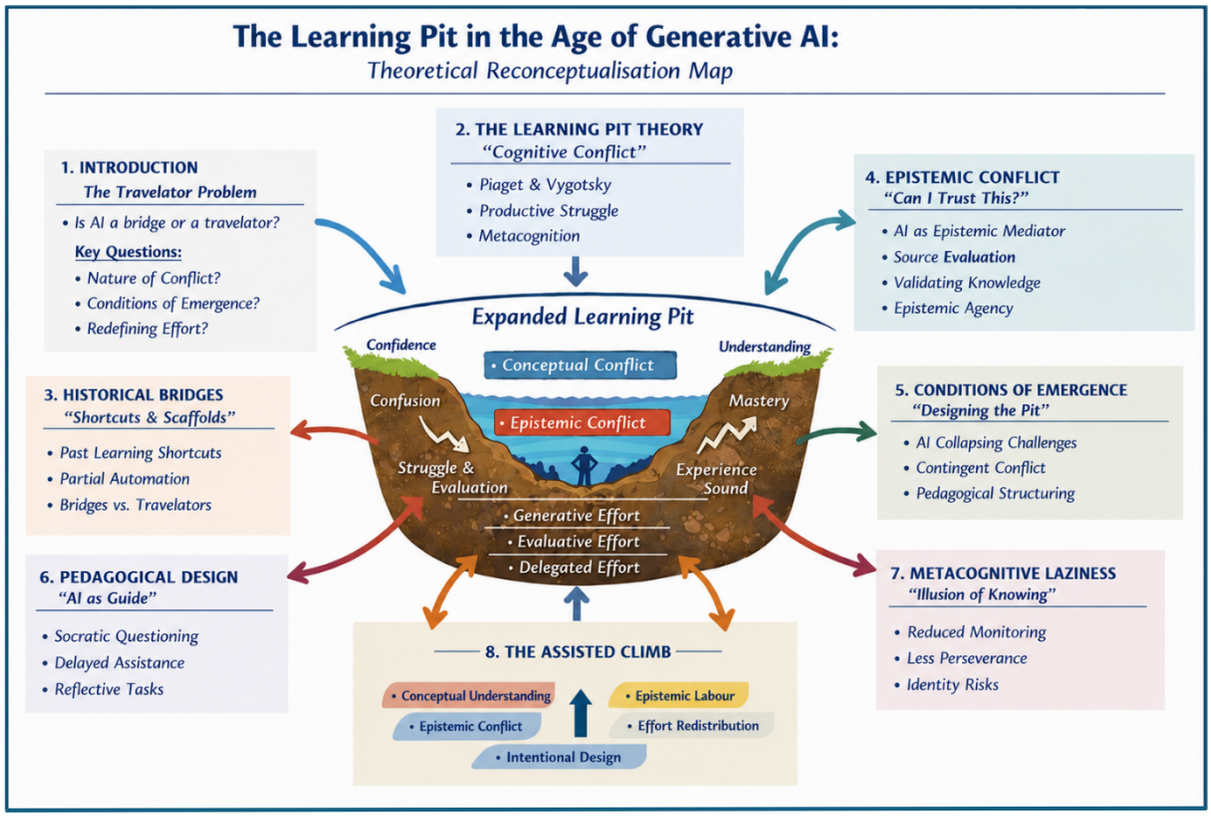


Figure 2. The Learning Pit in the Age of Generative AI.

The challenge is therefore no longer whether AI supports learning, but how human and artificial agents should share epistemic labour (Edwards, 2005) while preserving learner agency (Johnson-Laird, 1980). Responding to this challenge, this paper introduces The Assisted Climb (Figure 3), a conceptual framework for human–AI co-agency that reconceptualises the Learning Pit for educational environments characterised by artificial agency. Rather than positioning AI as a source of answers, the framework models learning as a co-agentic process in which humans and AI contribute differently to inquiry while learners retain epistemic responsibility.

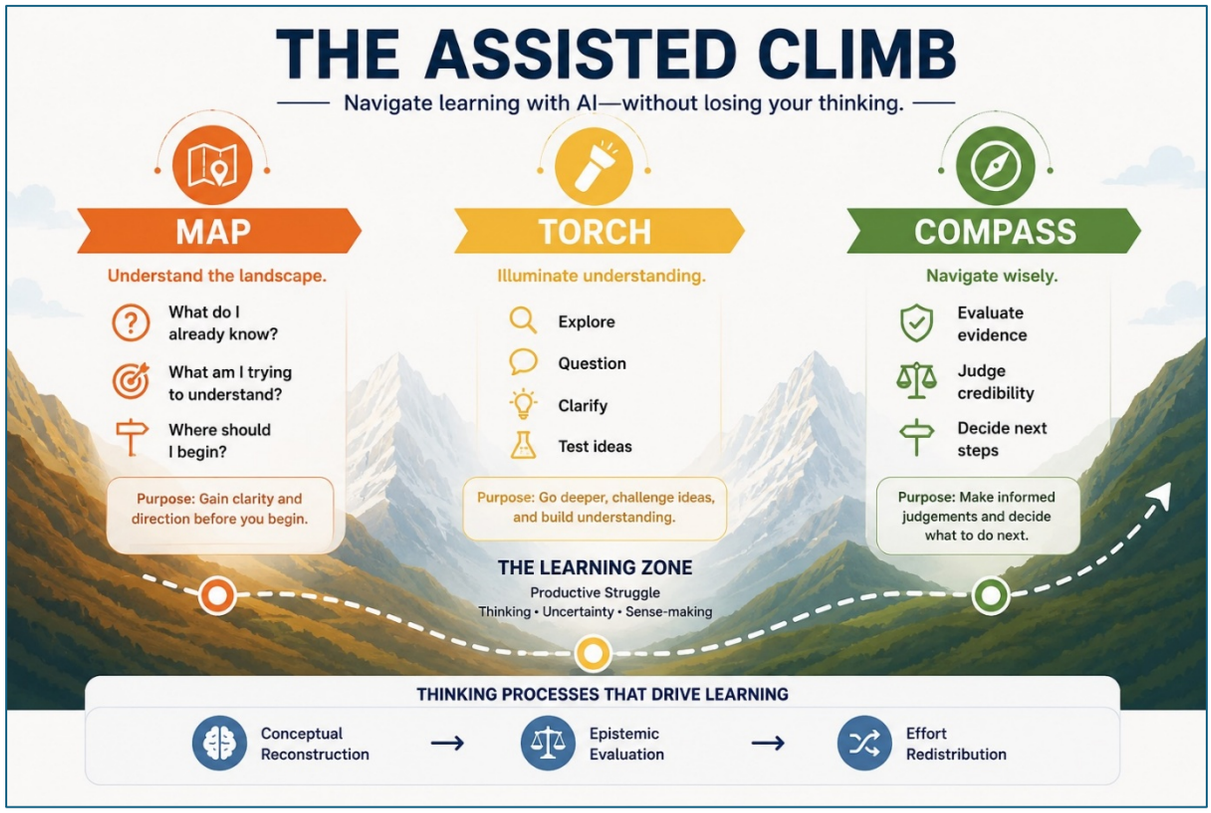


Figure 3. The Assisted Climb Model.

Research Aim, Question and Rationale

The aim of this study is to investigate how the Assisted Climb framework supports productive human–AI co-agency during an immersive learning experience involving Year 9 students in an Oxfordshire secondary school. As agentic AI increasingly participates in problem framing, knowledge generation, and evaluation, the study explores whether the framework enables learners to engage critically with AI while retaining epistemic responsibility for learning. The pilot serves as an initial empirical exploration of how the Assisted Climb mediates the redistribution of epistemic labour between human and artificial agents within authentic educational contexts. The central research question guiding this study is:

To what extent does the Assisted Climb framework support productive human–AI co-agency by enabling Year 9 learners to negotiate conceptual and epistemic conflict while maintaining epistemic agency during an immersive learning experience?

The rationale for this research is twofold. First, the emergence of agentic AI creates an urgent need for pedagogical frameworks that move beyond viewing AI as a tool towards

understanding how humans and AI can participate in learning as co-agents while preserving human epistemic agency. As AI increasingly contributes to reasoning, explanation, and evaluation, educators require approaches that help learners regulate AI participation rather than simply use AI effectively. Second, although the Assisted Climb is conceptually grounded in theories of cognitive conflict and socially mediated learning, its practical value as a framework for human–AI co-agency requires empirical investigation within authentic classroom settings. By examining how learners negotiate conceptual struggle, epistemic labour, and AI participation, this study positions the Assisted Climb as more than a reconceptualisation of the Learning Pit. Instead, it offers an emerging pedagogical framework for learning with agentic AI that seeks to preserve human agency while recognising AI as an increasingly active participant in knowledge construction.

How The Assisted Climb Was Developed

The Assisted Climb emerged from recognising that the challenge posed by agentic AI is not simply that learners have access to more powerful tools, but that the nature of learning itself is changing. The Learning Pit (Nottingham, 2020) assumes that learners undertake the intellectual work of resolving conceptual conflict, supported by teachers and resources. While education has always involved balancing depth with efficiency (Chan & Hu, 2023), earlier forms of support, including scaffolding, worked examples, and solution manuals, typically reduced the difficulty of learning without fundamentally redistributing responsibility for knowledge construction (Piaget, 1952; Vygotsky, 1978; Nottingham & Nottingham, 2017). Learners still crossed the Learning Pit through their own conceptual reconstruction, even when supported by others (Nottingham & Larsson, 2018).

Agentic AI in contrast represents a qualitatively different educational phenomenon as rather than simply assisting learners, increasingly autonomous AI systems can participate in problem framing, knowledge generation, explanation, and evaluation (Abdelghani et al., 2023). As a consequence, epistemic labour is redistributed across human and artificial agents (Floridi & Nobre, 2024), shifting AI from an instructional aid to an active participant in learning (Adel, 2024) and without guidance, there is a risk of an AI-heavy weighting within a human-AI learning pairing. The challenge is therefore no longer whether AI supports cognition, but how learners regulate this emerging relationship while retaining epistemic responsibility.

This redistribution of epistemic labour transforms the nature of learning. Conceptual conflict remains essential when existing ideas are challenged (Jose et al., 2025; Lin & Hung, 2025), but it is increasingly accompanied by epistemic conflict as learners must evaluate AI-generated knowledge that appears fluent yet lacks transparent reasoning (Floridi & Chiriatti, 2020). Learning therefore extends beyond determining what is correct to judging what is credible, when AI should be trusted, and where human judgement must prevail (Yan et al., 2025). Without these regulatory processes, cognitive offloading may become cognitive substitution, increasing the risk of epistemic drift (Hollan et al., 2026). These conditions require more than conventional AI literacy (Ng et al., 2021), instead they require pedagogies that explicitly support human–AI co-agency, enabling learners to regulate how epistemic labour is shared while maintaining ownership of inquiry. Building on concepts of metaliteracy (Mackey & Jacobson, 2011), the Assisted Climb conceptualises this as Generative AI Meta-literacy, in which learners monitor and regulate cognition distributed across human and AI agents rather than focusing solely on their own thinking (Shapiro et al., 2026).

The Assisted Climb was developed as a pedagogical response to these changing conditions. Rather than treating AI as a mechanism for reducing effort, the framework assumes that agentic AI redistributes cognitive and epistemic work, requiring learners to actively regulate AI participation while preserving human epistemic agency. To support this process, the model introduces three complementary regulatory functions: Map, Torch, and Compass, that structure productive human–AI co-agency.

- The Map represents the evolving knowledge landscape co-produced through human and AI contributions. It provides orientation rather than certainty, requiring learners to interpret, interrogate, and refine AI-generated knowledge because it remains probabilistic and incomplete (Floridi, 2025; Stamatellos, 2025).
- The Torch represents active inquiry where learners illuminate uncertainty by questioning AI outputs, refining prompts, seeking alternative perspectives, and sustaining conceptual struggle despite AI's persuasive fluency (Abdelghani et al., 2023; Yusuf et al., 2024).

- The Compass represents epistemic regulation because it enables learners to judge what to trust, when to challenge AI, and how to integrate human and AI contributions without relinquishing responsibility for knowledge construction (Fan et al., 2025).

Together, these functions reconceptualise the Learning Pit as a framework for educational environments characterised by artificial agency. Rather than removing struggle, the Assisted Climb positions learning as a co-agentic process in which humans and AI contribute differently to knowledge construction while learners retain epistemic responsibility. In doing so, it provides a pedagogical framework for negotiating conceptual conflict, regulating epistemic labour, and sustaining meaningful human agency within increasingly agentic learning environments.

Methodology

This study adopted a mixed-methods pre–post intervention design to investigate how the Assisted Climb framework supports productive human–AI co-agency during an immersive learning experience. Using a theory-testing interpretative case study approach (Klein & Myers, 1999; Johansson, 2007), the study examined how learners navigated conceptual and epistemic conflict while working alongside agentic AI. Rather than evaluating AI as an instructional tool, it explored how learners distributed, regulated, and retained epistemic labour during these interactions (Floridi & Nobre, 2024). To capture these processes, quantitative survey data were triangulated with ethnographic observations and analysis of student workbooks, enabling examination of both changes in learners' perceptions and the ways they regulated AI participation (Jose et al., 2025).

Participants and Recruitment

Participants were 41 Year 9 students (aged 13–14) from a secondary school in Oxfordshire, United Kingdom. The intervention formed part of the school's normal curriculum, with one complete cohort participating during an off-timetable enrichment day. All students completed the full six-hour programme. Ethical approval was granted by the required Research Ethics Committee and all procedures complied with institutional and school ethical requirements.

Instrument and Procedure

The intervention was designed to progressively develop learners' capacity for productive human–AI co-agency. Students first completed a pre-intervention survey exploring their existing understanding of AI, patterns of AI use, perceptions of themselves as learners, and familiarity with key concepts. They then completed an initial "Big Question" inquiry using Google AI Overview (the only AI not blocked on the school's internal server), providing a baseline of how they naturally interacted with AI prior to explicit instruction. Researchers observed these interactions to identify patterns of cognitive offloading, trust, prompting behaviours, and epistemic regulation. Students subsequently participated in four sequential lessons:

1. Why does learning sometimes feel difficult? – introducing cognitive conflict and productive struggle as foundations for conceptual change (Piaget, 1952; Nottingham, 2020).
2. Who am I as a learner? – exploring learner identity, metacognition, and self-regulation (Zimmerman, 2002).
3. Can AI always be trusted? – examining epistemic uncertainty, AI fluency, and the evaluation of AI-generated knowledge (Sharples, 2023).
4. When I use AI, how can I remain in the pilot seat? – introducing the Assisted Climb and its three regulatory functions: Map, Torch, and Compass as mechanisms for regulating human–AI co-agency (Floridi, 2025).

The final lesson explicitly addressed the redistribution of epistemic labour within human–AI systems (Humphreys, 2025), encouraging learners to interrogate, challenge, and regulate AI contributions rather than accept them uncritically. Following this instruction, students completed a second Big Question inquiry with Google AI Overview enabling comparison of their interactions before and after engagement with the Assisted Climb framework. The intervention concluded with a post-intervention survey measuring changes in understanding, confidence, and conceptual vocabulary. Throughout the day, researchers conducted structured ethnographic observations while students completed four workbooks containing written reflections, diagrams, prompts, and explanations of their reasoning. Collectively, these data provided insight into how learners regulated AI participation, negotiated

epistemic labour, and maintained ownership of inquiry while working alongside AI (Yan et al., 2025).

Data Triangulation

Due to the nature of human–AI co-agency as simultaneously behavioural, cognitive, and epistemic, no single source of evidence could adequately capture how learners negotiated AI participation. The study therefore employed data triangulation (Denzin, 1978) as shown in Figure 4 to examine the phenomenon from complementary perspectives, strengthening the credibility and interpretive validity of the findings (Noble & Heale, 2019). This approach was particularly appropriate given that reasoning may be distributed across learners and AI systems, making internal cognitive processes difficult to infer from any single data source (Kojukhov et al., 2025).

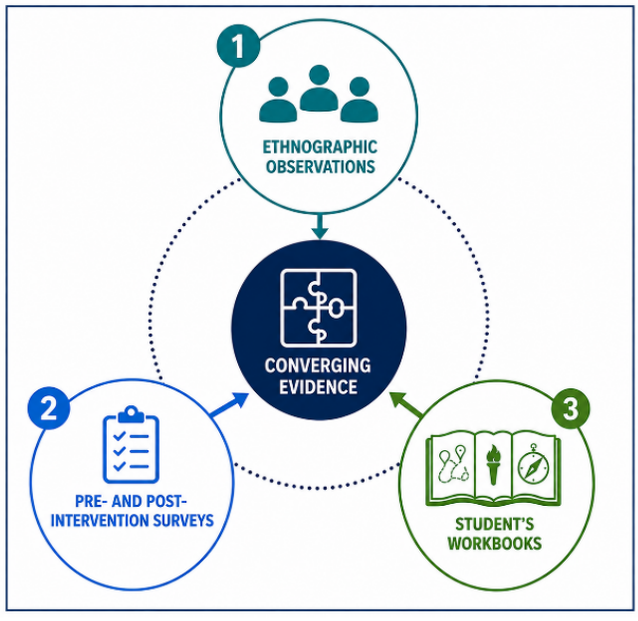


Figure 4. Data Triangulation.

The three complementary datasets were analysed included ethnographic observations which captured real-time interactions with AI, including prompting strategies, questioning behaviours, hesitation, and moments of uncritical acceptance or challenge, providing evidence of how learners regulated AI participation (Lin & Hung, 2025). Pre- and post-intervention surveys examined changes in learners' confidence, understanding, and perceptions of learning with AI following engagement with the Assisted Climb. Student workbooks documented learners' evolving reasoning, decision-making, and application of the Map, Torch, and Compass, providing evidence of how conceptual understanding and

epistemic regulation developed across the intervention (Floridi, 2025). Triangulating these datasets enabled the study to examine human–AI co-agency across behavioural, conceptual, epistemic, and attitudinal dimensions (Niemi et al., 2022). Rather than focusing solely on learning outcomes, the methodology captured how learners negotiated the redistribution of epistemic labour while maintaining responsibility for knowledge construction within increasingly agentic learning environments.

Data

To examine how the intervention influenced students' conceptualisation and enactment of human–AI co-agency, data from the observations, questionnaires and workbook activities were analysed separately before being compared across methods. Due to parent/carer consent for activity participation but not data sharing, although there were 41 students present, only a maximum of 25 individual data points were collected and compared, except for 23 for the post-intervention survey due to time constraints.

Observational Data

Table 1 presents the frequency of observable behaviours associated with human–AI interaction during the pre-intervention learning activity. The observations capture how students initially engaged with generative AI, including prompting behaviours, use of verification strategies, reliance on prior knowledge, and the extent to which students combined their own thinking with AI-generated responses.

Table 1. Quantitative indicators of co-agency (pre-intervention)

Indicator	Frequency	% of cohort
Copied teacher question directly into AI	Majority (estimated >30)	>73%*
Broke question into smaller prompts independently	0	0%
Required teacher prompting before decomposing question	Majority	—
Accepted first AI response without challenge	Majority	—
Requested sources/evidence	0	0%
Cross-checked AI with another AI	1	2.4%
Combined own ideas with AI output	3	7.3%
Asked AI to simplify language	3	7.3%
Declared "I've got my answer, I'm done"	4	9.8%
Rewrote AI output in own words	0	0%
Explicitly challenged AI accuracy	0	0%
Drew upon prior knowledge while interacting with AI	~5	12.2%

Table 2 summarises the observable behaviours recorded during the post-intervention activity. Rather than reporting individual frequencies, the table documents the occurrence of key behaviours associated with collaborative human–AI interaction across student groups, including prompt planning, iterative dialogue, evaluation of AI outputs, verification practices, and knowledge synthesis.

Table 2. Quantitative indicators (post-intervention)

Indicator	Observation
Students hesitant after first prompt	0 observed
Students asking, "What do I do now?"	0 observed
Students decomposing questions	Common across groups
Students planning prompts collaboratively	Common across groups
Students evaluating language	Frequently observed
Students verifying sources	Frequently observed
Students comparing perspectives	Frequently observed
Students using Google to verify AI	Observed
Students questioning AI truthfulness	Multiple groups observed
Students challenging AI using prior knowledge	At least 1 explicit observation
Students synthesising multiple ideas into final response	Commonly observed

Table 3 provides a comparative overview of changes in observable patterns of agency between the pre- and post-intervention activities. The matrix contrasts the relative prominence of behaviours relating to learner control, AI influence, interaction style, verification, and the application of human judgement.

Table 3. Agency Shift Matrix

Behaviour	Pre	Post
AI determines thinking	High	Low
Student determines thinking	Low	High
Single prompt interaction	High	Low
Iterative interaction	Low	High
Human judgement applied	Low	High
AI accepted as authority	High	Low
Human knowledge valued	Low	High
Verification behaviour	Very low	High

Table 4 presents a structured observation index developed to compare the presence or absence of key behaviours associated with human–AI co-agency across the two phases of the intervention. The indicators represent observable actions related to prompt planning, iterative interaction, critical evaluation, knowledge integration, and learner decision-making.

Table 4. A Co-agency Observation Index

Observable behaviour	Pre	Post
Student plans prompt before submission	✗	✓
Student revises prompt	✗	✓
Student asks follow-up questions	Rare	Frequent
Student challenges AI	✗	✓
Student verifies AI	Rare	Frequent
Student contributes own knowledge	Rare	Frequent
Student synthesises AI with own thinking	Rare	Frequent
Student treats AI as final authority	Frequent	Rare
Student exercises judgement	Rare	Frequent

Survey Data

Table 5 summarises the content analysis of students' open-ended questionnaire responses collected prior to the intervention (N = 25). Responses were coded inductively into themes describing students' perceptions of AI-supported learning, with frequencies and percentages reported for each code.

Table 5. Pre-intervention open-ended survey responses

Content code	n	% of respondents
AI primarily provides answers, information or completes tasks	16	64.0
AI can support understanding or learning	7	28.0
Learning depends on how AI is used (conditional learning)	8	32.0
Human remains responsible / in control when using AI	5	20.0
Verification or critical evaluation of AI outputs	5	20.0
AI should complete work or thinking for the learner (substitution model)	6	24.0

Table 6 presents representative quotations illustrating each of the themes identified within the pre-intervention questionnaire responses. The quotations provide examples of how students described the role of AI in learning before participating in the intervention.

Table 6. Illustrative quotations supporting the emergent pre-intervention themes

Theme	Representative student quotations
AI as an answer provider or task completer	"Answering specific questions."; "Giving answers straight away."; "Fast answers."; "Searching and helping you."; "Doing long work."; "Everything."; "Homework."
AI supporting learning or understanding	"To help understand concepts."; "Build study guides from my notes."; "Explain things to you."; "Helping you learn new things."; "Help for prep."
Learning depends on how AI is used	"It depends if it's copying it word for word."; "If they are using it to help them understand a concept, then it's learning."; "Partially because you are still learning but getting help."; "Yes and no."; "Depends."
Human remains responsible / in control	"I'm the one in control asking the questions."; "I'm in control."; "The controller."; "To ask questions."; "To make sure it is unbiased and accurate."
Verification / critical evaluation	"Check the citations."; "Look at the sources."; "Own knowledge."; "Research."; "Ask a different AI."
Substitution model of AI use	"Type my question and let AI do everything else."; "Use it for everything."; "To find answers."; "Mostly for homework."; "Answer simple questions."

Table 7 summarises the content analysis of students' post-intervention questionnaire responses relating to human–AI co-agency (N = 23). Responses were coded into thematic categories describing students' conceptions of AI's role within learning, with frequencies and percentages reported for each category.

Table 7. Post-intervention open-ended survey responses

Content code	n	% of respondents
AI should support learning rather than replace thinking	15	65.2
Human should remain in control ("pilot seat")	14	60.9
AI should explain, scaffold or support understanding	9	39.1
Learning depends on how AI is used	8	34.8
Critical evaluation or verification of AI outputs	4	17.4
AI should provide answers or complete work (substitution model)	3	13.0

Table 8 presents representative quotations illustrating the themes identified in the post-intervention responses. These excerpts demonstrate how students described the relationship between learner responsibility and AI support following completion of the intervention.

Table 8. Illustrative quotations supporting the emergent post-intervention themes

Theme	Representative student quotations
AI should support learning rather than replace thinking	"If used correctly."; "Help understand concepts."; "Help with explanations but not the whole questions."; "Use AI to learn but not for completion."; "They are as long as they are using it for the goal of learning and not completion."
AI as a learning scaffold	"Help explain hard to understand concepts."; "As a help if I'm stuck."; "With my research."; "To create study methods useful to me."; "Just ask it to explain."
Maintaining human agency ('pilot seat')	"You're in charge not the AI."; "Remain in control."; "Still use your own brain."; "Don't let AI do all the work."; "Do lots of the thinking yourself."; "Make sure to do the learning myself."
Learning depends on how AI is used	"Depends how they use it."; "As part, yes. As whole, no."; "If they take in the knowledge."; "Learning and not completion."
Critical AI literacy	"Checking AI accuracy and sources."; "Fact check."; "AI isn't reliable."; "Don't rely on it."
Substitution model of AI use	"Just ask the question and copy it."; "Do the homework."; "Ask for the answers."

Table 9 compares the proportion of respondents associated with each thematic category before and after the intervention. The comparison provides an overview of changes in students' reported conceptions of AI-supported learning across the two survey administrations.

Table 9. Pre- and post-intervention thematic category responses

Theme	Pre	Post
AI supports learning rather than replaces it	28.0%	65.2%
Human remains in control	20.0%	60.9%
AI as explanation/scaffold	28.0%	39.1%
Learning depends on how AI is used	32.0%	34.8%
Critical evaluation	20.0%	17.4%
AI as task completion/substitution	24.0%	13.0%

Workbook Data

The workbook activities were analysed to examine how students externalised their thinking while planning and regulating AI-supported learning. Table 10 summarises evidence from *The Map* activity, reporting the frequency with which key indicators of human–AI co-agency were evident within students' completed workbook artefacts (n = 25).

Table 10. Evidence of human–AI co-agency from *The Map* activity

Theme	Evidence from pupil artefacts	Frequency
Human ownership of problem framing	Pupils independently identified key concepts, prior knowledge, knowledge gaps and mission statements before AI use.	**22/25 (88%)**
Planning before AI interaction	All pupils completed workflow plans before prompting AI, demonstrating that task organisation preceded AI engagement.	**25/25 (100%)**
Allocation of cognitive responsibility	Pupils explicitly assigned different stages of the task to either themselves or AI, with more workflow steps allocated to humans (≈58%) than AI (≈42%).	**25/25 (100%)**
Distinguishing AI support from AI replacement	Pupils differentiated between AI supporting learning (e.g., explanations, examples, idea generation) and replacing learning (e.g., giving answers or completing the task).	**23/25 (92%)**
Human monitoring and evaluation	Pupils identified the need to question, verify information, check sources or make final decisions following AI use.	**21/25 (81%)**

Table 11 summarises responses from the *Trust Scale* workbook activity (n = 25). The table presents qualitative themes alongside associated quantitative evidence describing how students calibrated trust in AI across different contexts, including perceptions of verification, human responsibility, and appropriate uses of AI during learning.

Table 11. Evidence of calibrated trust and co-agency from the *Trust Scale* activity

Theme	Evidence from pupil responses	Quantitative evidence
Context-dependent trust	Pupils varied trust according to task rather than expressing uniformly high or low trust.	Evident across all 25 responses
AI as learning support	Brainstorming, explaining concepts and generating revision questions attracted the highest trust ratings.	Mean ≈ **6.2/10**
Low trust for high-stakes decisions	Medical and legal advice consistently received the lowest trust ratings.	Mean ≈ **3.5/10**
Human responsibility for verification	Pupils emphasised checking information, verifying sources and recognising AI errors.	**17/25 (68%)** explicitly referenced verification or checking
Humans remain decision-makers	Pupils stated that humans should remain in control when decisions involve health, law or significant consequences.	**18/25 (72%)** explicitly referred to human control or responsibility
Reduced cognitive offloading	AI was trusted less for writing essays than for supporting learning, suggesting pupils retained ownership of knowledge construction.	**22/25 (88%)** rated essay writing ≤6/10

Findings as a Structured Discussion

Consistent with the study's theory-testing mixed-methods design, this section is presented as a structured discussion rather than a conventional results section. As the study seeks to evaluate, rather than generate, theory, the discussion is organised around the core propositions of the Assisted Climb, examining how the integrated findings support, refine, or challenge its assumptions. Rather than presenting datasets separately, evidence is synthesised to explore changes in students' perceptions of AI, engagement with conceptual struggle, and patterns of human–AI co-agency, enabling systematic evaluation of the intervention's explanatory value.

Redistributed Epistemic Labour

The principle finding of this study is that not only did the Assisted Climb intervention improve students' ability to use generative AI; it reshaped how epistemic labour (AI (Latour, 2005) was shared between human and while simultaneously retaining human responsibility for learning (Floridi & Nobre, 2024). When comparing the three datasets at the start and end point of the intervention, observations, workbook responses, and pre- vs post-intervention surveys demonstrated students gained an appreciation that AI should be viewed as a bounded cognitive partner to extend, rather than replace, their own thinking (MacLeod et. al., 2019).

Prior to the intervention, students engaged with AI in a linear manner, with most copying the researcher's question directly into the system, accepting the initial response with little interrogation, request for evidence or compared information with alternative sources. Several students were observed explicitly regarding AI's first response as the completion of the task, suggesting that responsibility for cognitive load and critical thinking resided with the AI (Gnanaprakasam & Lourdusamy, 2024). Following the intervention however, students were observed displaying markedly different patterns of engagement and rather than accepting initial AI output, they compared AI-generated information with external sources, peer discussions and their own prior knowledge. Importantly, students no longer treated AI as the principal source of epistemic authority (Stamatellos, 2025) and instead, they increasingly viewed their own understanding as the reference point against which AI-generated information should be evaluated (Floridi, 2025).

The information gathered in student workbooks reinforced these observations, especially the Map task, in which students consistently identified learning goals, activated prior knowledge and recognised gaps in understanding before interacting with AI. Workflow planning similarly positioned AI after the initial stages of inquiry, assigning it bounded functions such as explanation, clarification and idea generation (Sundar, 2020) while reserving planning, organisation, evaluation and final decision-making for the learner (Floridi & Nobre, 2024). Prompt-planning activities further demonstrated that students increasingly sought explanations and alternative perspectives rather than completed answers, suggesting that AI was being incorporated into inquiry rather than replacing it (Kang & Lou, 2022).

Students' survey responses also reflected a comparable change in behaviour, with 65.2% of participants explicitly stating that AI should support learning without replacing thinking, while 60.9% independently referred to maintaining human control, frequently describing themselves as remaining "in the pilot seat." AI was primarily viewed as a scaffold for explanation, research and support when learners became stuck (39.1%), whereas only 13.0% continued to describe AI principally as a means of completing work. Although relatively few students (17.4%) explicitly discussed verification, this suggests that understanding the principle of human responsibility may develop before sophisticated evaluative behaviours become fully established (Stamatellos, 2025).

The convergence across all three datasets suggests that rather than learning to produce better prompts alone, students appeared to reconceptualise AI as one contributor within a wider process of knowledge construction (Latour, 2005) in which responsibility for judgement remained fundamentally human (Floridi & Nobre, 2024). These findings contribute to debates surrounding cognitive offloading (Hutchins, 2001), human–AI collaboration (MacLeod et al., 2019; Floridi & Nobre, 2024) and the risk that generative AI reduces productive cognitive struggle by automating aspects of reasoning traditionally undertaken by learners (Floridi & Nobre, 2024; Gnanaprakasam & Lourdusamy, 2024). The present findings also suggest a more nuanced interpretation because cognitive effort was not eliminated but redistributed (Hutchins, 2001). Students appeared to devote less effort to information retrieval and more effort to planning inquiry, evaluating evidence, integrating perspectives and exercising

epistemic judgement. This supports emerging arguments that AI redistributes, rather than replaces, epistemic labour (MacLeod et al., 2019), while extending theories of distributed cognition (Hutchins, 2001) by demonstrating that educationally productive collaboration depends upon explicit regulation of human responsibility (Floridi & Nobre, 2024). It can be thus argued the Assisted Climb shifts attention from teaching students how to use AI efficiently towards helping them determine which aspects of inquiry should remain under human control within increasingly agentic learning environments (Floridi, 2025; Stamatellos, 2025).

Epistemic Workflow

A second finding was that the Assisted Climb intervention changes how students understood the process of learning with AI. Rather than simply acquiring better prompting techniques, learners developed a structured epistemic workflow (Paul & Elder, 2007; Edwards, 2005) in which planning, interaction, evaluation and synthesis became distinct stages of AI-supported inquiry (MacLeod et al., 2019). Before the intervention, many students lacked a coherent strategy for engaging with AI and researcher observations frequently noted uncertainty following AI's first response, with students asking how to continue the conversation or deepen the information provided. AI interactions typically followed a simple ask-and-answer pattern in which responses were copied or briefly summarised with minimal refinement, comparison or critical evaluation. Few students analysed the task before consulting AI or considered how AI outputs might be integrated into their own developing understanding (Sundar, 2020).

Following the intervention, these observed behaviours changed with students increasingly approaching AI as part of an iterative process rather than a single exchange (Kang & Lou, 2022). Before opening an AI, they analysed the task, identified learning goals and reformulated broad questions into smaller, purposeful enquiries. During interaction they refined prompts, interrogated explanations, compared alternative viewpoints and sought clarification where responses appeared incomplete or ambiguous. After AI interaction, learners increasingly evaluated information against other sources, peer discussion and prior knowledge before constructing their own conclusions. The earlier uncertainty regarding "what happens next?" disappeared almost entirely, suggesting that students had internalised

a more coherent model of AI-supported inquiry (Edwards, 2005; MacLeod et al., 2019; Floridi & Nobre, 2024).

Workbook evidence provides further support for this interpretation with student responses in planning activities consistently demonstrating that they regarded inquiry design as a human responsibility (Edwards, 2005; Floridi & Nobre, 2024), with AI introduced only after learning goals and conceptual understanding had been established. Workflow tasks similarly located AI within a wider sequence of planning, explanation, evaluation and synthesis rather than at the beginning or end of learning. Prompt-planning exercises reflected this shift, with students increasingly designing prompts that requested clarification, explanation or alternative perspectives rather than complete answers. Several students also explicitly incorporated verification and source-checking into their workflows, indicating an emerging understanding that productive AI use requires ongoing epistemic regulation (Floridi, 2025; Kang & Lou, 2022).

Survey responses reflected similar conceptual developments with 65.2% of respondents describing AI as supporting rather than replacing learning as well as 34.8% explicitly arguing that educational value depended upon how AI was used rather than AI itself. In this sense, they students described AI as helping them understand difficult concepts, support research or overcome obstacles, positioning it within an ongoing learning process rather than as a substitute for thinking (MacLeod et al., 2019; Sundar, 2020). 13% of substitution-oriented responses suggests that most participants had moved beyond viewing AI primarily as an answer-generation system, although the persistence of these responses also indicates that changes in epistemic practice may require sustained reinforcement beyond a single intervention (Edwards, 2005).

Taken together, these findings challenge the growing emphasis on prompt engineering as the principal indicator of AI competence (Sundar, 2020). Prompting emerged as only one element within a broader cycle of planning, questioning, evaluation and synthesis (Paul & Elder, 2007; Edwards, 2005). This aligns with theories of metacognitive regulation (Paul & Elder, 2007) while extending them into the context of human–AI collaboration (MacLeod et al., 2019; Floridi & Nobre, 2024), where learners must monitor not only their own cognition but also

the quality, reliability and contribution of AI-generated knowledge (Floridi, 2025; Kang & Lou, 2022). Within the Assisted Climb, prompting becomes subordinate to inquiry rather than replacing it. Effective AI literacy therefore depends less upon asking increasingly sophisticated questions and more upon developing increasingly sophisticated models of how inquiry should unfold across human and artificial contributors (Edwards, 2005; Latour, 2005; Floridi, 2025).

Calibrated Epistemic Trust

A third finding was that the intervention transformed students' understanding of trust in AI with learners increasingly demonstrating calibrated epistemic trust (Stamatellos, 2025), in which confidence in AI depended on the nature of the task, the potential consequences of error and the continuing involvement of human judgement (Floridi, 2025). Trust therefore became less a property of AI itself and more a judgement about when and how AI should contribute to learning (Kaplan & Haenlein, 2019; MacLeod et al., 2019).

Before the intervention, researcher observations suggested that students exhibited implicit blanket trust. AI responses were typically accepted without challenge, requests for evidence or justification were rare and few learners compared AI's outputs with alternative sources. Several students regarded AI's first response as sufficient to complete the task, indicating that AI was frequently treated as an authoritative source (Stamatellos, 2025) rather than one requiring critical evaluation (Floridi & Nobre, 2024). Following the intervention, students were observed adopting verification as a routine component of inquiry (Kang & Lou, 2022). They questioned explanations, refined prompts, compared information across AI systems, internet searches, teachers and peers, and in several instances deliberately sought evidence from recognised organisations before accepting AI-generated claims. Importantly, students also began to draw on their own prior knowledge when evaluating AI outputs, demonstrating that epistemic authority (Edwards, 2005; Latour, 2005) had shifted from AI towards a negotiated relationship in which human judgement remained central.

Workbook activities revealed a similarly nuanced understanding of trust. Rather than expressing consistently high or low confidence in AI, students differentiated between tasks according to their epistemic demands (Paul & Elder, 2007; Floridi, 2025). AI was generally

considered appropriate for explanation, brainstorming, revision and idea generation but substantially less appropriate for medical, legal or other high-stakes decisions where accuracy and accountability were paramount (Behdadi & Munthe, 2020). Students consistently justified these distinctions in terms of responsibility rather than technical capability, recognising that AI could contribute valuable support while humans retained accountability for consequential decisions (Behdadi & Munthe, 2020).

Survey responses reinforced these findings with 60.9% in the post-intervention survey stating that humans should remain "in charge" or "in the pilot seat" throughout AI-supported inquiry. 39.1% described AI as helping learners understand concepts or overcome difficulties, while 13% continued to advocate substitution-oriented uses such as completing homework or generating final answers. Although only 17.4% of students discussed verification, this suggests that conceptual recognition of human responsibility may precede the development of a sophisticated vocabulary for describing verification practices (Floridi, 2025).

Collectively, these findings indicate that students moved beyond viewing trust as a simple attitude towards AI. Instead, they increasingly understood trust as a continuous process of deciding when AI should contribute (Floridi & Nobre, 2024), what forms of support were appropriate and where human responsibility should remain non-negotiable (Kaplan & Haenlein, 2019). This extends current discussions of AI trust by suggesting that educationally productive trust is neither unquestioning acceptance nor blanket scepticism, but an ongoing process of epistemic regulation (Floridi, 2025; Kang & Lou, 2022). Verification therefore becomes not evidence of distrust but a defining characteristic of effective human–AI collaboration (MacLeod et al., 2019). At the same time, the persistence of a small number of substitution-oriented responses suggests that calibrated trust is unlikely to emerge through isolated interventions alone and instead requires sustained pedagogical reinforcement across multiple learning contexts (Floridi & Nobre, 2024).

Human–AI Co-Agency

It can be thus argued these findings provide a pedagogical operationalisation of human–AI co-agency (Floridi, 2025) by making explicit how epistemic labour (Edwards, 2005) should be negotiated between learners and increasingly agentic AI systems (MacLeod et al., 2019).

Across observations, workbook activities and survey responses, students demonstrated changes not only in how they used AI but also in how they understood agency, responsibility and knowledge construction. Following the intervention, rather than treating AI as either a passive tool or an autonomous authority (Floridi & Nobre, 2024; Stamatellos, 2025), students appeared to develop an explicit understanding that productive learning depended upon negotiated participation (Edwards, 2005) in which different forms of cognitive work were intentionally distributed (Hutchins, 2001) while epistemic authority remained fundamentally human (Stamatellos, 2025).

This pattern was evident across all datasets and suggests that agency was not transferred from AI back to humans (Floridi & Nobre, 2024), rather, agency became distributed across the interaction while remaining asymmetrical (Dung, 2024). AI contributed increasingly sophisticated explanations, ideas and information, but learners retained intentionality, evaluative authority and accountability throughout inquiry (Floridi, 2025; Stamatellos, 2025). This distinction is theoretically important because it challenges binary accounts that portray AI either as a neutral educational tool or as an autonomous educational actor (Floridi & Nobre, 2024; Stamatellos, 2025). Instead, the findings support a relational understanding of agency (Müller & Schurr, 2016) in which learning emerges through structured collaboration between human and artificial contributors (MacLeod et al., 2019; Dung, 2024) whose responsibilities remain differentiated rather than equal (Floridi & Nobre, 2024).

In the Assisted Climb: the Map encourages learners to frame problems, activate prior knowledge and determine where AI may contribute before interaction begins; the Torch supports iterative inquiry by encouraging learners to question, refine and extend AI-generated knowledge rather than accepting initial responses uncritically and; the Compass guides verification, evaluation and judgement, ensuring that responsibility for deciding what should ultimately be accepted remains with the learner. Together, these elements extend theoretical discussions of distributed cognition (Hutchins, 2001) and human–AI co-agency (Dung, 2024; Floridi, 2025) by demonstrating that co-agency need not remain a philosophical description of AI-supported learning but can instead be intentionally cultivated through explicit pedagogical design. Rather than focusing primarily on prompt engineering or technical AI competence (Sundar, 2020), the Assisted Climb foregrounds epistemic

governance (Floridi, 2025): helping learners determine what should be delegated to AI, what should remain under human control and how responsibility should be negotiated throughout inquiry (Floridi & Nobre, 2024).

Limitations and Future Research

Despite these findings, it worth noting this study should be interpreted within several limitations, primarily as it represents a small-scale pilot involving 41 Year 9 students from a single secondary school. Although convergence across observations, workbook artefacts and survey responses strengthens the credibility of the findings (Guba & Lincoln, 1994), the study was designed to explore the pedagogical potential of the Assisted Climb rather than produce generalisable claims (Yin, 2003). The intervention comprised a single six-hour learning experience and although students demonstrated shifts in their conceptualisation of AI, epistemic responsibility (Floridi & Nobre, 2024) and interaction patterns, it is unclear whether these changes would persist or transfer into everyday practice. Longitudinal research should examine whether co-agentic behaviours become embedded through sustained engagement with AI (MacLeod et al., 2019; Dung, 2024; Floridi, 2025). The study also examined learners' regulation of epistemic labour (Hutchins, 2001) rather than academic attainment. Future research should therefore investigate whether shifts towards human–AI co-agency (Dung, 2024) improve disciplinary understanding, critical thinking, metacognitive regulation (Paul & Elder, 2007) and longer-term learning outcomes.

As agentic AI continues to evolve, the Assisted Climb should also be evaluated across different large language models and increasingly agentic systems. Research should also examine how teachers negotiate epistemic responsibility (Edwards, 2005; Floridi & Nobre, 2024) when designing AI-supported learning, as co-agency depends on both learner and teacher decision-making (Edwards, 2005; MacLeod et al., 2019). The findings also suggest that human–AI co-agency (Edwards, 2005; Latour, 2005; Dung, 2024; Floridi, 2025) is better understood as a developmental disposition than a competence acquired through a single intervention. Future work should therefore move beyond prompt engineering and AI literacy (Kaplan & Haenlein, 2019; Sundar, 2020) towards pedagogies that preserve human epistemic

agency (Floridi & Nobre, 2024; Stamatellos, 2025) while supporting productive collaboration with increasingly capable artificial agents (Dung, 2024; Floridi, 2025).

Conclusion

The emergence of increasingly agentic AI represents far more than the introduction of another educational technology. As AI systems become capable of contributing to problem framing, knowledge generation, and evaluation, they fundamentally reshape how epistemic labour is distributed within learning, shifting the central educational challenge from whether AI should be used to how it can participate in learning while preserving human responsibility for judgement, intentionality, and knowledge construction. This study argues that meeting this challenge requires moving beyond viewing AI as either a passive instructional tool or an autonomous educational actor towards understanding learning as a process of human–AI co-agency. The Assisted Climb extends Nottingham's Learning Pit by recognising that productive struggle in the age of agentic AI is both conceptual and epistemic, requiring learners to negotiate what should be delegated to AI, what should remain under human control, and how responsibility should be shared throughout inquiry. Across classroom observations, workbook activities, and student surveys, the findings indicate that the framework supported meaningful shifts in students' relationships with AI, with learners retaining responsibility for planning, evaluation, and decision-making, developing structured workflows that positioned AI within rather than in place of inquiry, demonstrating more calibrated and context-sensitive trust, and conceptualising AI as a bounded cognitive partner rather than an answer-generation system.

Together, these findings suggest that epistemic labour can be redistributed without diminishing learner agency. The principal contribution of this study is therefore both theoretical and pedagogical: rather than describing human–AI co-agency as a philosophical ideal, the Assisted Climb provides an initial pedagogical framework through which co-agency can be intentionally cultivated, redirecting attention from prompt engineering and technical competence towards the regulation of epistemic relationships between learners and increasingly agentic AI. As artificial agency continues to evolve, educational success will depend not simply on learners' ability to use AI effectively, but on their capacity to make informed decisions about what should be delegated, what should remain uniquely human,

and how responsibility for knowledge construction should be negotiated within collaborative human–AI systems. In this respect, the Assisted Climb offers an important first step towards a pedagogy that enables learners to work alongside increasingly capable AI while remaining the primary authors of their own learning.

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Figure 1. The Learning Pit. (Nottingham, 2017)

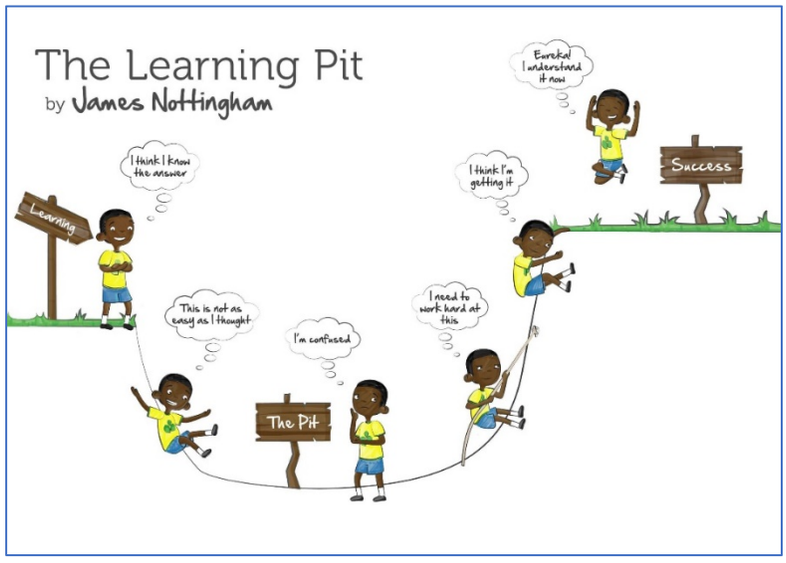


Figure 2. The Learning Pit in the Age of Generative AI.

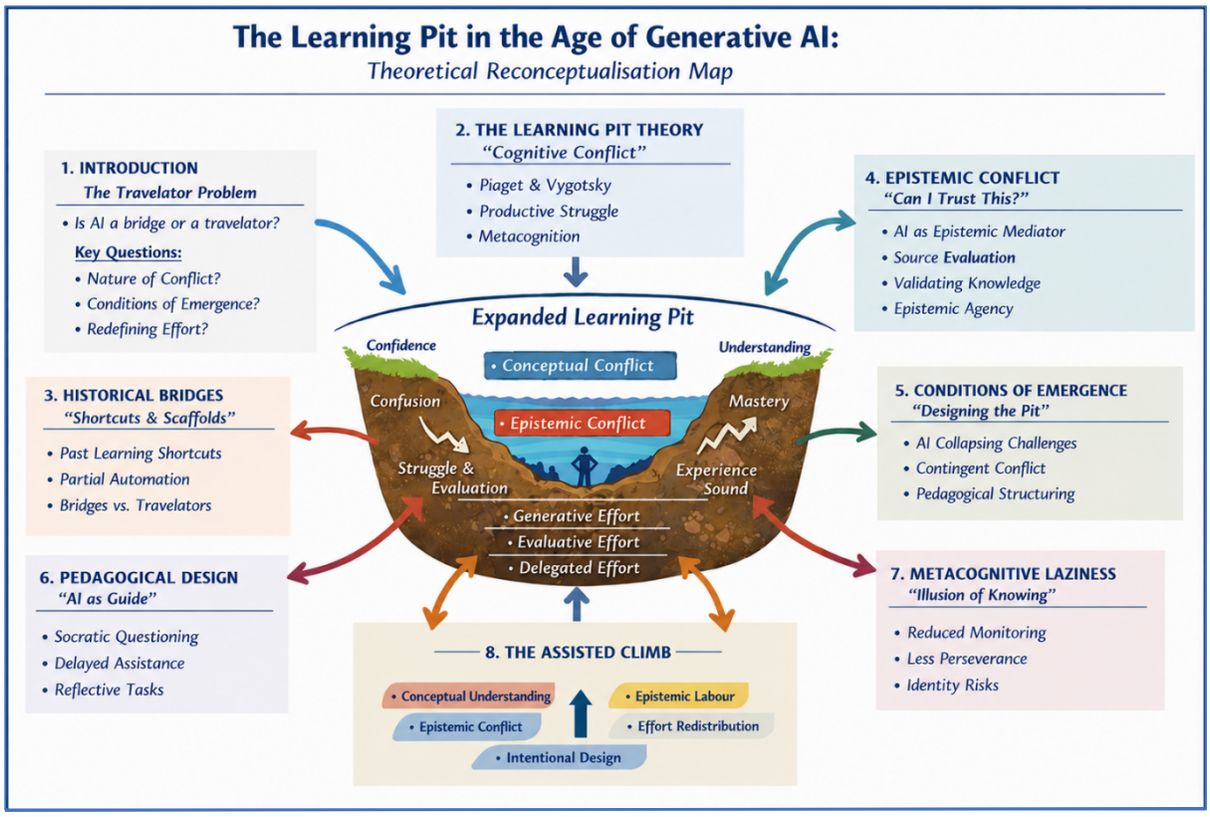


Figure 3. The Learning Pit in the AI Age.

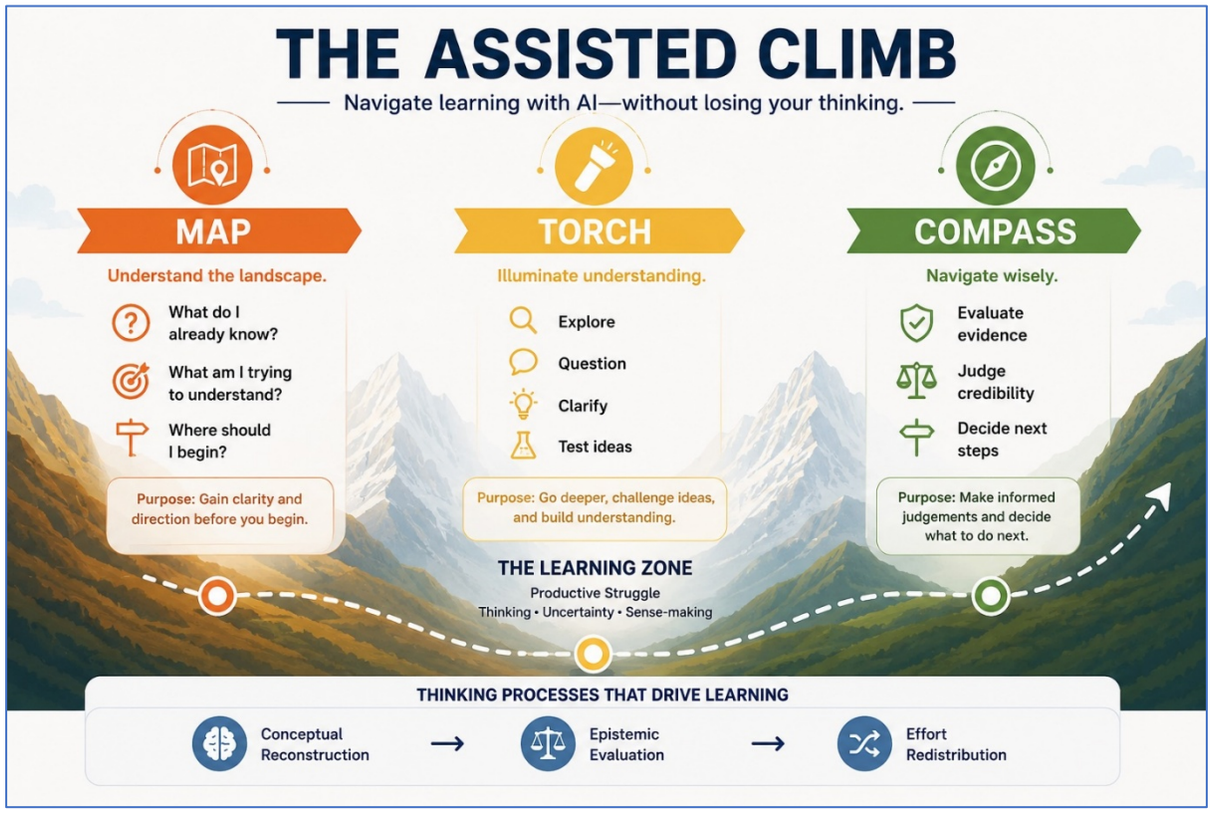
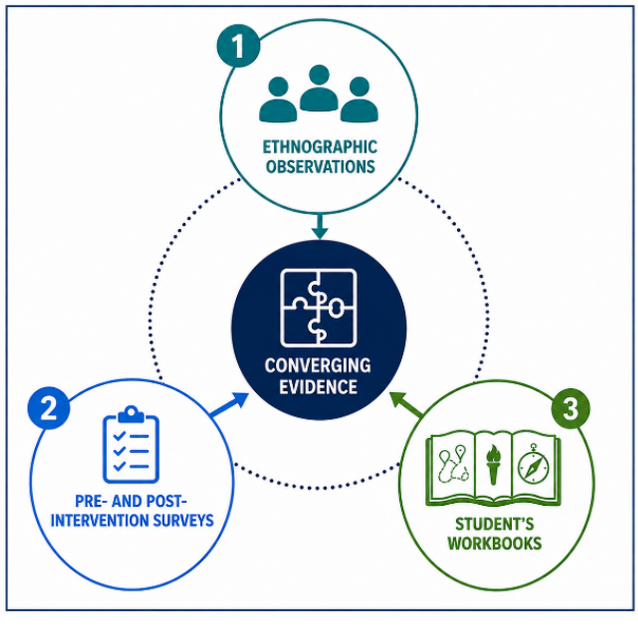


Figure 4. Data Triangulation.



Indicator	Frequency	% of cohort
Copied teacher question directly into AI	Majority (estimated >30)	>73%*
Broke question into smaller prompts independently	0	0%
Required teacher prompting before decomposing question	Majority	—
Accepted first AI response without challenge	Majority	—
Requested sources/evidence	0	0%
Cross-checked AI with another AI	1	2.4%
Combined own ideas with AI output	3	7.3%
Asked AI to simplify language	3	7.3%
Declared "I've got my answer, I'm done"	4	9.8%
Rewrote AI output in own words	0	0%
Explicitly challenged AI accuracy	0	0%
Drew upon prior knowledge while interacting with AI	~5	12.2%

Table 1. Quantitative indicators of co-agency (pre-intervention)

Indicator	Observation
Students hesitant after first prompt	0 observed
Students asking, "What do I do now?"	0 observed
Students decomposing questions	Common across groups
Students planning prompts collaboratively	Common across groups
Students evaluating language	Frequently observed
Students verifying sources	Frequently observed
Students comparing perspectives	Frequently observed
Students using Google to verify AI	Observed
Students questioning AI truthfulness	Multiple groups observed
Students challenging AI using prior knowledge	At least 1 explicit observation
Students synthesising multiple ideas into final response	Commonly observed

Table 2. Quantitative indicators (post-intervention)

Behaviour	Pre	Post
AI determines thinking	High	Low
Student determines thinking	Low	High
Single prompt interaction	High	Low
Iterative interaction	Low	High
Human judgement applied	Low	High
AI accepted as authority	High	Low
Human knowledge valued	Low	High
Verification behaviour	Very low	High

Table 3. Agency Shift Matrix

Observable behaviour	Pre	Post
Student plans prompt before submission	✗	✓
Student revises prompt	✗	✓
Student asks follow-up questions	Rare	Frequent
Student challenges AI	✗	✓
Student verifies AI	Rare	Frequent
Student contributes own knowledge	Rare	Frequent
Student synthesises AI with own thinking	Rare	Frequent
Student treats AI as final authority	Frequent	Rare
Student exercises judgement	Rare	Frequent

Table 4. A Co-agency Observation Index

Content code	n	% of respondents
AI primarily provides answers, information or completes tasks	16	64.0
AI can support understanding or learning	7	28.0
Learning depends on how AI is used (conditional learning)	8	32.0
Human remains responsible / in control when using AI	5	20.0
Verification or critical evaluation of AI outputs	5	20.0
AI should complete work or thinking for the learner (substitution model)	6	24.0

Table 5. Pre-intervention open-ended survey responses

Theme	Representative student quotations
AI as an answer provider or task completer	"Answering specific questions."; "Giving answers straight away."; "Fast answers."; "Searching and helping you."; "Doing long work."; "Everything."; "Homework."
AI supporting learning or understanding	"To help understand concepts."; "Build study guides from my notes."; "Explain things to you."; "Helping you learn new things."; "Help for prep."
Learning depends on how AI is used	"It depends if it's copying it word for word."; "If they are using it to help them understand a concept, then it's learning."; "Partially because you are still learning but getting help."; "Yes and no."; "Depends."
Human remains responsible / in control	"I'm the one in control asking the questions."; "I'm in control."; "The controller."; "To ask questions."; "To make sure it is unbiased and accurate."
Verification / critical evaluation	"Check the citations."; "Look at the sources."; "Own knowledge."; "Research."; "Ask a different AI."
Substitution model of AI use	"Type my question and let AI do everything else."; "Use it for everything."; "To find answers."; "Mostly for homework."; "Answer simple questions."

Table 6. Illustrative quotations supporting the emergent pre-intervention themes

Content code	n	% of respondents
AI should support learning rather than replace thinking	15	65.2
Human should remain in control ("pilot seat")	14	60.9
AI should explain, scaffold or support understanding	9	39.1
Learning depends on how AI is used	8	34.8
Critical evaluation or verification of AI outputs	4	17.4
AI should provide answers or complete work (substitution model)	3	13.0

Table 7. Post-intervention open-ended survey responses

Theme	Representative student quotations
AI should support learning rather than replace thinking	"If used correctly."; "Help understand concepts."; "Help with explanations but not the whole questions."; "Use AI to learn but not for completion."; "They are as long as they are using it for the goal of learning and not completion."
AI as a learning scaffold	"Help explain hard to understand concepts."; "As a help if I'm stuck."; "With my research."; "To create study methods useful to me."; "Just ask it to explain."
Maintaining human agency ('pilot seat')	"You're in charge not the AI."; "Remain in control."; "Still use your own brain."; "Don't let AI do all the work."; "Do lots of the thinking yourself."; "Make sure to do the learning myself."
Learning depends on how AI is used	"Depends how they use it."; "As part, yes. As whole, no."; "If they take in the knowledge."; "Learning and not completion."
Critical AI literacy	"Checking AI accuracy and sources."; "Fact check."; "AI isn't reliable."; "Don't rely on it."
Substitution model of AI use	"Just ask the question and copy it."; "Do the homework."; "Ask for the answers."

Table 8. Illustrative quotations supporting the emergent post-intervention themes

Theme	Pre	Post
AI supports learning rather than replaces it	28.0%	65.2%
Human remains in control	20.0%	60.9%
AI as explanation/scaffold	28.0%	39.1%
Learning depends on how AI is used	32.0%	34.8%
Critical evaluation	20.0%	17.4%
AI as task completion/substitution	24.0%	13.0%

Table 9. Pre- and post-intervention thematic category responses

Theme	Evidence from pupil artefacts	Frequency
Human ownership of problem framing	Pupils independently identified key concepts, prior knowledge, knowledge gaps and mission statements before AI use.	**22/25 (88%)**
Planning before AI interaction	All pupils completed workflow plans before prompting AI, demonstrating that task organisation preceded AI engagement.	**25/25 (100%)**
Allocation of cognitive responsibility	Pupils explicitly assigned different stages of the task to either themselves or AI, with more workflow steps allocated to humans (≈58%) than AI (≈42%).	**25/25 (100%)**
Distinguishing AI support from AI replacement	Pupils differentiated between AI supporting learning (e.g., explanations, examples, idea generation) and replacing learning (e.g., giving answers or completing the task).	**23/25 (92%)**
Human monitoring and evaluation	Pupils identified the need to question, verify information, check sources or make final decisions following AI use.	**21/25 (81%)**

Table 10. Evidence of human–AI co-agency from *The Map* activity

Theme	Evidence from pupil responses	Quantitative evidence
Context-dependent trust	Pupils varied trust according to task rather than expressing uniformly high or low trust.	Evident across all 25 responses
AI as learning support	Brainstorming, explaining concepts and generating revision questions attracted the highest trust ratings.	Mean ≈ **6.2/10**
Low trust for high-stakes decisions	Medical and legal advice consistently received the lowest trust ratings.	Mean ≈ **3.5/10**
Human responsibility for verification	Pupils emphasised checking information, verifying sources and recognising AI errors.	**17/25 (68%)** explicitly referenced verification or checking
Humans remain decision-makers	Pupils stated that humans should remain in control when decisions involve health, law or significant consequences.	**18/25 (72%)** explicitly referred to human control or responsibility
Reduced cognitive offloading	AI was trusted less for writing essays than for supporting learning, suggesting pupils retained ownership of knowledge construction.	**22/25 (88%)** rated essay writing ≤6/10

Table 11. Evidence of calibrated trust and co-agency from the *Trust Scale* activity